

Flumazenil: Drug information - UpToDate

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For additional information see "Flumazenil: Patient drug information" and "Flumazenil: Pediatric drug information"

For abbreviations, symbols, and age group definitions show table

ALERT: US Boxed Warning

Seizures:

The use of flumazenil has been associated with the occurrence of seizures. These are most frequent in patients who have been on benzodiazepines for long-term sedation or in overdose cases where patients are showing signs of serious cyclic antidepressant overdose. Practitioners should individualize the dosage of flumazenil and be prepared to manage seizures.

Pharmacologic Category

- Antidote

Dosing: Adult

Benzodiazepine reversal, procedural sedation or general anesthesia

Benzodiazepine reversal, procedural sedation or general anesthesia:

IV: Initial: 0.2 mg over 2 minutes; if the desired level of consciousness is not obtained after 1 minute, 0.2 mg may be repeated at 1-minute intervals up to 4 times; usual cumulative dosage range: 0.6 to 1 mg; maximum cumulative dose: 1 mg (Ref).

Resedation: IV: In the event of resedation, repeat 0.2 to 1 mg doses (administered at 0.1 mg/minute) may be administered at 20-minute intervals as needed; maximum cumulative resedation dose: 3 mg in 1 hour (Ref).

Benzodiazepine overdose, treatment

Benzodiazepine overdose, treatment: Note: Routine use of flumazenil is controversial, as most benzodiazepine-poisoned patients can be treated successfully with supportive measures alone, and use of flumazenil is associated with a risk of adverse effects (eg, withdrawal seizures, arrhythmias), especially in patients who have co-ingested a proconvulsant or proarrhythmic agent, patients that are physiologically dependent on benzodiazepines, or patients with an underlying seizure disorder (Ref). Consider consultation with poison control center or medical toxicologist; refer to institutional protocols.

IV: Initial: 0.2 mg over 2 minutes; if the desired level of consciousness is not obtained 30 seconds after the dose, 0.3 mg can be administered over 3 minutes; if the desired level of consciousness is still not obtained, 0.5 mg can be administered over 5 minutes and repeated at 1-minute intervals; usual cumulative dosage range: 1 to 3 mg; usual cumulative maximum dose: 3 mg (Ref).

Note: Patients with a partial response to a cumulative dose of 3 mg may rarely require additional titration up to a total dose of 5 mg (although doses >3 mg do not reliably produce additional effects). If a patient has not responded 5 minutes after a cumulative dose of 5 mg, benzodiazepines are likely not the major cause of sedation; sedation may be due to exposure to additional CNS depressants (eg, opioids).

Resedation: IV: In the event of resedation, repeat 0.5 to 1 mg doses (administered at 0.1 mg/minute) may be administered at 20-minute intervals as needed; maximum cumulative resedation dose: 3 mg in 1 hour (Ref).

Note: After an overdose with a large dose of a benzodiazepine, the duration of a single dose of flumazenil is not expected to exceed 1 hour; if clinically necessary, the period of wakefulness may be prolonged with repeated low IV doses of flumazenil, or by an infusion of 0.1 to 1 mg/hour (off label) (Ref). Use of a continuous infusion of flumazenil has been associated with a reduction in the number of patients requiring intubation as compared to use of supportive care alone (Ref); consider consultation with a poison control center or medical toxicologist when selecting flumazenil dosing strategy.

Nonbenzodiazepine hypnotic overdose or gabapentin overdose, treatment

Nonbenzodiazepine hypnotic overdose or gabapentin overdose, treatment (off-label use):

Note: May consider during the treatment of patients who are experiencing toxicity secondary to a nonbenzodiazepine hypnotic agent (eg, eszopiclone, zaleplon, zolpidem, zopiclone) or gabapentin (Ref). Limited data available; dose based on use in benzodiazepine overdose.

IV: Initial: 0.2 mg over 2 minutes; if the desired level of consciousness is not obtained 30 seconds after the dose, 0.3 mg can be administered over 3 minutes; if the desired level of consciousness is still not obtained, 0.5 mg can be administered over 5 minutes and repeated at 1-minute intervals; usual cumulative dosage range: 1 to 3 mg; usual maximum cumulative dose: 3 mg (Ref).

Note: Based on use in benzodiazepine overdose, patients with a partial response to a cumulative dose of 3 mg may rarely require additional titration up to a total dose of 5 mg (although doses >3 mg do not reliably produce additional effects). If a patient has not responded 5 minutes after a cumulative dose of 5 mg, the major cause of sedation may be due to exposure to additional CNS depressants (eg, opioids).

Resedation: IV: In the event of resedation, repeat 0.5 to 1 mg doses (administered at 0.1 mg/minute) may be administered at 20-minute intervals if needed; maximum cumulative resedation dose: 3 mg in 1 hour (Ref).

Note: Due to the short duration of sedation of nonbenzodiazepine hypnotics, a continuous infusion is probably unnecessary in patients intoxicated with these agents (Ref).

Dosing: Kidney Impairment: Adult

The renal dosing recommendations are based upon the best available evidence and clinical expertise. Senior Editorial Team: Bruce Mueller, PharmD, FCCP, FASN, FNKF; Jason A. Roberts, PhD, BPharm (Hons), B App Sc, FSHP, FISAC; Michael Heung, MD, MS.

Altered kidney function: IV: No dosage adjustment necessary for any degree of kidney impairment (Ref).

Hemodialysis, intermittent (thrice weekly): IV: Not significantly dialyzed: No supplemental dose or dosage adjustment necessary (Ref).

Peritoneal dialysis: IV: Not significantly dialyzed: No dosage adjustment necessary (Ref).

CRRT: IV: No dosage adjustment necessary (Ref).

PIRRT: IV: No dosage adjustment necessary (Ref).

Dosing: Liver Impairment: Adult

Initial reversal: No dosage adjustment necessary. Repeat doses: Reduce dose or frequency.

Dosing: Older Adult

Refer to adult dosing. No differences in safety or efficacy have been reported; however, increased sensitivity may occur in some elderly patients.

Dosing: Pediatric

(For additional information see "Flumazenil: Pediatric drug information")

Benzodiazepine intoxication/overdose

Benzodiazepine intoxication/overdose: Limited data available:

Note: Consult a clinical toxicologist or poison control center for guidance on patient selection for flumazenil therapy. The role of flumazenil in the poisoned patient, especially comatose patients and mixed overdose patients, remains controversial (Ref). Flumazenil should not be used with unknown ingestions or in patients with any of the following: chronic maintenance with or physiologic dependence on benzodiazepines, receiving a benzodiazepine to treat a life-threatening condition, coingestion of a proconvulsant (eg, tricyclic antidepressants), or history of an underlying seizure disorder (Ref).

Infants, Children, and Adolescents:

IV: Initial dose: 0.01 mg/kg; maximum dose: 0.2 mg/dose; may repeat every minute if needed to a maximum cumulative dose of 1 mg total (Ref).

Continuous IV infusion: 0.005 to 0.01 mg/kg/**hour** has been reported as an alternative to intermittent doses (Ref).

Benzodiazepine reversal, procedural sedation or general anesthesia

Benzodiazepine reversal, procedural sedation or general anesthesia: Infants, Children, and Adolescents: IV: Initial dose: 0.01 mg/kg; maximum dose: 0.2 mg/dose; given over 15 seconds; if needed, may repeat same dose after 45 seconds, and then every minute to a maximum cumulative dose of 0.05 mg/kg or 1 mg total, whichever is lower; usual total dose: 0.08 to 1 mg (mean: 0.65 mg) (Ref).

Dosing: Kidney Impairment: Pediatric

There are no dosage adjustments provided in the manufacturer's labeling; adult pharmacokinetic data suggest drug not significantly affected by renal failure (CrCl <10 mL/minute) or hemodialysis.

Dosing: Liver Impairment: Pediatric

Initial reversal dose: Use normal dose; repeat doses should be decreased in size or frequency.

Adverse Reactions

The following adverse drug reactions and incidences are derived from product labeling unless otherwise specified. Adverse reactions reported in children, adolescents, and adults.

>10%: Gastrointestinal: Vomiting (11%)

1% to 10%:

Cardiovascular: Flushing (1% to 3%), palpitations (3% to 9%), vasodilation (cutaneous) (1% to 3%)

Dermatologic: Diaphoresis (1% to 3%)

Endocrine & metabolic: Hot flash (1% to 3%)

Gastrointestinal: Nausea (1% to 3%), xerostomia (3% to 9%)

Local: Inflammation at injection site (1% to 3%), injection-site reaction (1% to 3%, including skin abnormality), pain at injection site (3% to 9%), rash at injection site (1% to 3%)

Nervous system: Agitation (3% to 9%), anxiety (3% to 9%), asthenia (1% to 3%), ataxia (10%), depersonalization (1% to 3%), depression (1% to 3%), dizziness (10%), dysphoria (1% to 3%), emotional lability (1% to 3%), euphoria (1% to 3%), fatigue (1% to 3%), headache (1% to 3%), hypoesthesia (1% to 3%), insomnia (3% to 9%), malaise (1% to 3%), nervousness (3% to 9%), paranoia ideation (1% to 3%), paresthesia (1% to 3%), sensation disorder (1% to 3%), tremor (3% to 9%), uncontrolled crying (1% to 3%), vertigo (10%)

Ophthalmic: Blurred vision (3% to 9%), diplopia (1% to 3%), lacrimation (1% to 3%), visual disturbance (1% to 3%), visual field defect (1% to 3%)

Respiratory: Dyspnea (3% to 9%), hyperventilation (3% to 9%)

<1%:

Cardiovascular: Atrial arrhythmia, atrioventricular nodal arrhythmia, bradycardia, cardiac arrhythmia, chest pain, hypertension, premature ventricular contractions, tachycardia, ventricular tachycardia

Gastrointestinal: Hiccups

Hypersensitivity: Tongue edema

Nervous system: Confusion, delirium, drowsiness, hyperacusis, lack of concentration, rigors, seizure, shivering, speech disturbance, stupor, voice disorder

Otic: Auditory disturbance, reversible hearing loss, tinnitus

Frequency not defined: Nervous system: Withdrawal syndrome

Postmarketing: Nervous system: Fear

Contraindications

Hypersensitivity to flumazenil, benzodiazepines, or any component of the formulation; patients administered benzodiazepines for control of potentially life-threatening conditions (eg, control of intracranial pressure or status epilepticus); patients who may have ingested or are showing signs of cyclic-antidepressant overdose.

Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Warnings/Precautions

Concerns related to adverse effects:

- Amnesia: Does not consistently reverse amnesia; patient may not recall verbal instructions after procedure.
- CNS depression: May cause CNS depression, which may impair physical or mental abilities; patients must be cautioned about performing tasks which require mental alertness (eg, operating machinery or driving) for 24 hours after discharge.
- Resedation: Occurs more frequently in patients where a large single dose or cumulative dose of a benzodiazepine has been administered along with a neuromuscular-blocking agent and multiple anesthetic agents.
- Respiratory depression: Should not rely upon to reverse respiratory depression/hypoventilation. Flumazenil is not a substitute for evaluation of oxygenation. Establishing an airway and assisting ventilation, as necessary, is always the initial step in overdose management.
- Seizures: **[US Boxed Warning]: Benzodiazepine reversal may result in seizures; seizures may occur more frequently in patients on benzodiazepines for long-term sedation or following tricyclic antidepressant overdose. Dose should be individualized and practitioners should be prepared to manage seizures.** Seizures may also develop in patients with concurrent

major sedative-hypnotic drug withdrawal, recent therapy with repeated doses of parenteral benzodiazepines, myoclonic jerking or seizure activity prior to flumazenil administration. Use with caution in patients relying on a benzodiazepine for seizure control.

Disease-related concerns:

- Head injury: Use with caution in patients with a head injury; may alter cerebral blood flow or precipitate convulsions in patients receiving benzodiazepines.
- Hepatic impairment: Use with caution in patients with hepatic dysfunction; repeated doses of the drug should be reduced in frequency or amount.
- Panic disorder: Use with caution in patients with a history of panic disorder; may provoke panic attacks.

Special populations:

- Drug/alcohol dependence: Use caution in drug and ethanol-dependent patients; these patients may also be dependent on benzodiazepines.
- Intensive care patients: Should be used with caution in the intensive care unit because of increased risk of unrecognized benzodiazepine dependence in such settings.

Other warnings/precautions:

- Appropriate use: Should not be used to diagnose benzodiazepine-induced sedation. Reverse neuromuscular blockade before considering use. Flumazenil does not antagonize the CNS effects of other GABA agonists (such as ethanol, barbiturates, or general anesthetics); nor does it reverse opioids. Not recommended for treatment of benzodiazepine dependence.
- Overdose use: Use with caution in patients with mixed drug overdoses; toxic effects of other drugs taken may emerge once benzodiazepine effects are reversed.

Dosage Forms: US

Excipient information presented when available (limited, particularly for generics); consult specific product labeling.

Solution, Intravenous:

Generic: 0.5 mg/5 mL (5 mL); 1 mg/10 mL (10 mL)

Generic Equivalent Available: US

Yes

Pricing: US

Solution (Flumazenil Intravenous)

0.5 mg/5 mL (per mL): \$1.56 - \$1.78

1 mg/10 mL (per mL): \$1.56 - \$9.35

Disclaimer: A representative AWP (Average Wholesale Price) price or price range is provided as reference price only. A range is provided when more than one manufacturer's AWP price is available and uses the low and high price reported by the manufacturers to determine the range. The pricing data should be used for benchmarking purposes only, and as such should not be used alone to set or adjudicate any prices for reimbursement or purchasing functions or considered to be an exact price for a single product and/or manufacturer. Medi-Span expressly disclaims all warranties of any kind or nature, whether express or implied, and assumes no liability with respect to accuracy of price or price range data published in its solutions. In no event shall Medi-Span be liable for special, indirect, incidental, or consequential damages arising from use of price or price range data. Pricing data is updated monthly.

Dosage Forms: Canada

Excipient information presented when available (limited, particularly for generics); consult specific product labeling.

Solution, Intravenous:

Generic: 0.1 mg/mL (5 mL)

Administration: Adult

IV: Administer in freely running IV into large vein to minimize pain at the injection site. To decrease frequency of emergent confusion and agitation, administer at a rate of 0.1 mg/minute, especially with larger doses. May also administer at a rate of 0.2 mg/minute or as fast as 0.2 mg over 15 seconds; however, there may be an increase in adverse effects (Ref).

Administration: Pediatric

Parenteral: For IV use only; administer by rapid IV injection over 15 to 30 seconds at a rate not to exceed 0.2 mg/minute; administer through a freely running IV infusion into larger vein (to decrease chance of pain, phlebitis).

Storage/Stability

Store at 20°C to 25°C (68°F to 77°F). Once drawn up in the syringe or mixed with D5W, LR, or NS, use within 24 hours. Discard any unused solution after 24 hours.

Pharmacy supply of emergency antidotes: Guidelines suggest that at least 6 to 12 mg of flumazenil be stocked. Suggested amount is stated to be a sufficient quantity to treat 1 patient weighing 100 kg for an initial 8- to 24-hour period (Dart 2018); actual amount to be stocked should take into account site-specific and population-specific needs.

Use: Labeled Indications

Benzodiazepine reversal, procedural sedation or general anesthesia: Complete or partial reversal of the sedative effects of benzodiazepines used in procedural sedation and general anesthesia.

Benzodiazepine overdose, treatment: Treatment of benzodiazepine overdose.

Use: Off-Label: Adult

Nonbenzodiazepine hypnotic overdose or gabapentin overdose, treatment

Medication Safety Issues

Sound-alike/look-alike issues:

Flumazenil may be confused with influenza virus vaccine

Romazicon may be confused with rocuronium

Metabolism/Transport Effects

None known.

Drug Interactions

(For additional information: Launch drug interactions program)

There are no known significant interactions.

Pregnancy Considerations

Teratogenic effects were not seen in animal reproduction studies. Embryocidal effects were seen at large doses. Use during labor and delivery is not recommended. In general, medications used as antidotes should take into consideration the health and prognosis of the mother; antidotes should be administered to pregnant women if there is a clear indication for use and should not be withheld because of fears of teratogenicity (Bailey 2003).

Breastfeeding Considerations

It is not known if flumazenil is excreted in breast milk. The manufacturer recommends that caution be used if administering to breast-feeding women.

Dietary Considerations

Avoid alcohol for the first 24 hours after administration or as long as the effects of benzodiazepines exist.

Monitoring Parameters

Monitor for return of sedation, respiratory depression, benzodiazepine withdrawal, seizures, and other residual effects of benzodiazepines for at least 2 hours and until the patient is stable and resedation is unlikely.

Mechanism of Action

Competitively inhibits the activity at the benzodiazepine receptor site on the GABA/benzodiazepine receptor complex. Flumazenil does not antagonize the CNS effect of drugs affecting GABA-ergic neurons by means other than the benzodiazepine receptor (ethanol, barbiturates, general anesthetics) and does not reverse the effects of opioids.

Pharmacokinetics (Adult Data Unless Noted)

Note: Follows a 2 compartment open model; Clearance and V_d per kg are similar for children and adults, but children display more variability.

Onset of action: 1 to 2 minutes; 80% response within 3 minutes.

Peak effect: 6 to 10 minutes.

Duration: Resedation may occur after ~1 hour (range: 19-50 minutes); duration related to dose administered and benzodiazepine plasma concentrations; reversal effects of flumazenil may wear off before effects of benzodiazepine.

Distribution: Initial V_d : 0.5 L/kg; V_{dss} : 0.9-1.1 L/kg.

Protein binding: ~50% (~67% of which is bound to albumin).

Metabolism: Hepatic; dependent upon hepatic blood flow.

Half-life elimination:

Children: Terminal: 20 to 75 minutes (mean: 40 minutes).

Adults: Alpha: 4 to 11 minutes; Terminal: 40 to 80 minutes.

Moderate hepatic dysfunction: 1.3 hours.

Severe hepatic impairment: 2.4 hours.

Excretion: Feces; urine (<1% as unchanged drug).

Clearance: Dependent upon hepatic blood flow; Adults: 0.8 to 1 L/hour/kg.

Pharmacokinetics: Additional Considerations (Adult Data Unless Noted)

Hepatic function impairment: **Moderate:** Mean total clearance decreased 40% to 60%. **Severe:** Mean total clearance decreased 75%.

Patient Counseling Points

(For additional information see "Flumazenil: Patient drug information")

What is this drug used for?

- It is used to reverse the effects of drugs called benzodiazepines (such as alprazolam, diazepam, and lorazepam). This includes a benzodiazepine overdose.

All drugs may cause side effects. However, many people have no side effects or only have minor side effects. Call your doctor or get medical help if any of these side effects or any other side effects bother you or do not go away:

- Dizziness or headache
- Upset stomach or throwing up
- Blurred eyesight
- Dry mouth
- Sweating a lot
- Feeling nervous and excitable
- Shakiness
- Trouble sleeping

WARNING/CAUTION: Even though it may be rare, some people may have very bad and sometimes deadly side effects when taking a drug. Tell your doctor or get medical help right away if you have any of the following signs or symptoms that may be related to a very bad side effect:

- Shortness of breath
- Fast breathing
- A heartbeat that does not feel normal
- Seizures
- Mood changes
- Change in how you act
- Feeling agitated
- Change in balance
- Redness, burning, pain, swelling, or leaking of fluid where the drug is going into the body
- Signs of an allergic reaction, like rash; hives; itching; red, swollen, blistered, or peeling skin with or without fever; wheezing; tightness in the chest or throat; trouble breathing, swallowing, or talking; unusual hoarseness; or swelling of the mouth, face, lips, tongue, or throat.

Note: This is not a comprehensive list of all side effects. Talk to your doctor if you have questions.

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Brand Names: International

International Brand Names by Country

For country code abbreviations (show table)

- (AE) United Arab Emirates: Anexate;
- (AR) Argentina: Flumazenil duncan | Flumazenil Surar | Lanexat;
- (AT) Austria: Anexate | Flumazenil B.Braun | Flumazenil deltaselect | Flumazenil Hikma | Flumazenil kabi | Flumazenil pharmaselect;
- (AU) Australia: Anexate | Apo flumazenil | DBL Flumazenil | Flumazenil kabi;
- (BE) Belgium: Anexate | Flumazenil b. braun;
- (BG) Bulgaria: Anexate | Flumazenil kabi;
- (BR) Brazil: Flumazil | Flunexil | Lanexat | Lenazen;
- (CH) Switzerland: Flumazenil Actavis | Flumazenil Labatec | Flumazenil mepha | Flumazenil Sintetica;
- (CL) Chile: Antadona | Lanexat;
- (CN) China: Anexate | Lai yi;
- (CO) Colombia: Diazenil | Lanexat;
- (CR) Costa Rica: Flumazil;
- (CZ) Czech Republic: Anexate | Flumazenil pharmaselect;
- (DE) Germany: Anexate | Flumazenil Actavis | Flumazenil b. braun | Flumazenil hexal | Flumazenil Inresa | Flumazenil pharmaselect | Flumazenil Teva;
- (DO) Dominican Republic: Lanexat;
- (EC) Ecuador: Lanexat;
- (EE) Estonia: Anexate | Flumazenil b. braun | Flumazenil fresenius kabi | Flumazenil pharmaselect | Mazenil;
- (ES) Spain: Anexate | Flumazenilo Actavis | Flumazenilo Braun | Flumazenilo Combino pharm | Flumazenilo fresenius kabi | Flumazenilo G.E.S. | Flumazenilo genfarma | Flumazenilo Hikma | Flumazenilo Teva;
- (FI) Finland: Flumazenil b braun | Lanexat;
- (FR) France: Anexate | Flumazenil Aguezzant | Flumazenil dakota | Flumazenil Hikma | Flumazenil kabi | Flumazenil panpharma | Flumazenil Teva | Flumazenil Winthrop;
- (GR) Greece: Anexate | Demoxate | Flumazenil/cooper | Flumazenil/kabi | Flumexat;
- (HU) Hungary: Anexate | Flumazenil kabi | Flumazenil pharmaselect;
- (IE) Ireland: Anexate;
- (IL) Israel: Anexate;
- (IN) India: Fludot;
- (IT) Italy: Anexate | Flumazenil kabi | Flumazenil Teva;
- (JO) Jordan: Exitine | Flunexate;
- (JP) Japan: Anexate | Flumazenil i.v. chemiphar | Flumazenil iv teva | Flumazenil sw;
- (KE) Kenya: Anexate | Exitine;
- (KR) Korea, Republic of: Anexate | Ansedative | Flunil | Lumasate | Luzenil | Newzenil;
- (KW) Kuwait: Anexate;
- (LB) Lebanon: Anexate;
- (LT) Lithuania: Anexate;
- (LV) Latvia: Anexate;
- (MA) Morocco: Anexate;
- (MX) Mexico: Antadona | Outnestin;
- (MY) Malaysia: Anexate | Antabenz | Flumazenil Hameln | Flumazenil kabi;
- (NL) Netherlands: Anexate | Flumazenil Actavis | Flumazenil kabi;
- (NO) Norway: Anexate | Flumazenil Actavis | Flumazenil aurora medical | Flumazenil b. braun | Flumazenil fresenius kabi | Flumazenil Hameln;
- (NZ) New Zealand: Anexate;
- (PE) Peru: Lanexat;
- (PL) Poland: Anexate | Flumazenil b. braun | Flumazenil kabi | Flumazenil pharmaselect;
- (PR) Puerto Rico: Romazicon;
- (PT) Portugal: Anexate;
- (PY) Paraguay: Lanexat;

- (QA) Qatar: Anesed-R | Anexate | Exitine | Mazenil;
- (RO) Romania: Anexate | Flumazenil pharmaselect;
- (RU) Russian Federation: Anexate;
- (SA) Saudi Arabia: Exitine;
- (SE) Sweden: Flumazenil Actavis | Flumazenil b. braun | Flumazenil fresenius kabi | Flumazenil Hameln | Lanexat;
- (SG) Singapore: Anexate;
- (SI) Slovenia: Anexate;
- (SK) Slovakia: Anexate;
- (SV) El Salvador: Flumazenil vijosa;
- (TN) Tunisia: Anexate | Demoxate | Flumazenil panpharma;
- (TR) Turkey: Anesed r | Anexate | Denaksat | Fluxate | Lauzenil | Mazenil;
- (TW) Taiwan: Anexate | Flumazenil Hameln;
- (UA) Ukraine: Flumazenil vista;
- (UG) Uganda: Exitine;
- (UY) Uruguay: Fadaflumaz | Flumazenilo icu vita | Flumil;
- (VE) Venezuela, Bolivarian Republic of: Flumazepin | Hemzenil | Lanexat;
- (ZA) South Africa: Anexate | Flumazenil fresenius kabi

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REFERENCES

1. An H, Godwin J. Flumazenil in benzodiazepine overdose. *CMAJ*. 2016;188(17-18):E537. doi:10.1503/cmaj.160357 [PubMed 27920113]
2. Bailey B. Are There Teratogenic Risks Associated With Antidotes Used in the Acute Management of Poisoned Pregnant Women? *Birth Defects Res A Clin Mol Teratol*. 2003, 67(2):133-140. [PubMed 12769509]
3. Burton JH, Lyon L, Dorfman T, Tomassoni AJ. Continuous flumazenil infusion in the treatment of zolpidem (Ambien) and ethanol coingestion. *J Toxicol Clin Toxicol*. 1998;36(7):743-746. doi:10.3109/15563659809162626 [PubMed 9865246]
4. Butler TC, Rosen RM, Wallace AL, Amsden GW. Flumazenil and dialysis for gabapentin-induced coma. *Ann Pharmacother*. 2003;37(1):74-76. doi:10.1345/aph.1C242 [PubMed 12503937]
5. Brammer G, Gilby R, Walter FG, et al. Continuous Intravenous Flumazenil Infusion for Benzodiazepine Poisoning. *Vet Hum Toxicol*. 2000;42(5):280-281. [PubMed 11003118]
6. Chern CH, Chern TL, Hu SC, et al. Complete and Partial Response to Flumazenil in Patients With Suspected Benzodiazepine Overdose. *Am J Emerg Med*. 1995;13(3):372-375. [PubMed 7755838]
7. Chern CH, Chern TL, Wang LM, et al. Continuous Flumazenil Infusion in Preventing Complications Arising From Severe Benzodiazepine Intoxication. *Am J Emerg Med*. 1998;16(3):238-241. [PubMed 9596422]
8. Cienki JJ, Burkhart KK, Donovan JW. Zopiclone overdose responsive to flumazenil. *Clin Toxicol (Phila)*. 2005;43(5):385-386. doi:10.1081/clt-200058944 [PubMed 16235515]
9. Clark RF, Sage TA, Tunget C, et al. Delayed Onset Lorazepam Poisoning Successfully Reversed By Flumazenil in a Child: Case Report and Review of the Literature. *Pediatr Emerg Care*. 1995;11(1):32-34. [PubMed 7739960]
10. Cone AM, Nadel S, Sweeney B. Flumazenil reverses diazepam-induced neonatal apnoea and hypotonia. *Eur J Pediatr*. 1993;152(5):458-459. doi:10.1007/BF01955914 [PubMed 8319721]
11. Dart RC, Goldfrank LR, Erstad BL, et al. Expert consensus guidelines for stocking of antidotes in hospitals that provide emergency care. *Ann Emerg Med*. 2018;71(3):314-325.e1. doi:10.1016/j.annemergmed.2017.05.021 [PubMed 28669553]

12. Dixon JC, Speidel BD, Dixon JJ. Neonatal Flumazenil Therapy Reverses Maternal Diazepam. *Acta Paediatr.* 1998;87(2):225-226. [PubMed 9512213]
13. Expert opinion. Senior Renal Editorial Team: Bruce Mueller, PharmD, FCCP, FASN, FNKF; Jason A. Roberts, PhD, BPharm (Hons), B App Sc, FSHP, FISAC; Michael Heung, MD, MS.
14. Flumazenil [prescribing information]. Deerfield, IL: Baxter Healthcare Corporation; August 2018.
15. Flumazenil [prescribing information]. Rockford, IL: Mylan; November 2013.
16. Ghouri AF, Ramirez Ruiz MA, White PF. Effect of Flumazenil on Recovery After Midazolam and Propofol Sedation. *Anesthesiology.* 1994;81(2):330-339.
17. Gunja N. The clinical and forensic toxicology of Z-drugs. *J Med Toxicol.* 2013;9(2):155-162. doi:10.1007/s13181-013-0292-0 [PubMed 23404347]
18. Hansen AR, Stark AR, Eichenwald EC, Martin CR. *Cloherty and Stark's Manual of Neonatal Care.* 9th ed. Wolters Kluwer Health; 2022.
19. Haverkos GP, Di Salvo RP, Imhoff TE. Fatal Seizures After Flumazenil Administration in a Patient With Mixed Overdose. *Ann Pharmacother.* 1994;28(12):1347-1349. [PubMed 7696723]
20. Höjer J, Baehrendtz S. The effect of flumazenil (Ro 15-1788) in the management of self-induced benzodiazepine poisoning. A double-blind controlled study. *Acta Med Scand.* 1988;224(4):357-365. doi:10.1111/j.0954-6820.1988.tb19595.x [PubMed 3142220]
21. Höjer J, Baehrendtz S, Magnusson A, et al. A placebo-controlled trial of flumazenil given by continuous infusion in severe benzodiazepine overdosage. *Acta Anaesthesiol Scand.* 1991;35(7):584-590. [PubMed 1686131]
22. Höjer J, Salmonson H, Sundin P. Zaleplon-induced coma and bluish-green urine: possible antidotal effect by flumazenil. *J Toxicol Clin Toxicol.* 2002;40(5):571-572. doi:10.1080/07313810.2002.11863667 [PubMed 12215053]
23. Hood SD, Norman A, Hince DA, Melichar JK, Hulse GK. Benzodiazepine dependence and its treatment with low dose flumazenil. *Br J Clin Pharmacol.* 2014;77(2):285-294. [PubMed 23126253]
24. Howland M. Flumazenil. In: Nelson LS, Howland M, Lewin NA, Smith SW, Goldfrank LR, Hoffman RS, eds. *Goldfrank's Toxicologic Emergencies.* 11th ed. McGraw-Hill Education; 2019.
25. Kreshak AA, Tomaszewski CA, Clark RF, Cantrell FL. Flumazenil administration in poisoned pediatric patients. *Pediatr Emerg Care.* 2012;28(5):448-450. doi:10.1097/PEC.0b013e3182531d0d [PubMed 22531190]
26. Lheureux P, Debailleul G, De Witte O, Askenasi R. Zolpidem intoxication mimicking narcotic overdose: response to flumazenil. *Hum Exp Toxicol.* 1990;9(2):105-107. doi:10.1177/096032719000900209 [PubMed 2111156]
27. Masciullo M, Pichiorri F, Scivoletto G, Foti C, Molinari M. Flumazenil therapy for a gabapentin-induced coma: a case report. *J Med Case Rep.* 2021;15(1):242. doi:10.1186/s13256-021-02816-3 [PubMed 33964989]
28. Mathieu-Nolf M, Babé MA, Coquelle-Couplet V, et al. Flumazenil Use in an Emergency Department: A Survey. *J Toxicol Clin Toxicol.* 2001;39(1):15-20. [PubMed 11327221]
29. Maxa JL, Ogu CC, Adeeko MA, Swaner TG. Continuous-infusion flumazenil in the management of chlordiazepoxide toxicity. *Pharmacotherapy.* 2003;23(11):1513-1516. doi:10.1592/phco.23.14.1513.31941 [PubMed 14620397]
30. Moore PW, Donovan JW, Burkhart KK, et al. Safety and efficacy of flumazenil for reversal of iatrogenic benzodiazepine-associated delirium/toxicity during treatment of alcohol withdrawal, a retrospective review at one center. *J Med Toxicol.* 2014;10(2):126-132. [PubMed 24619543]
31. Nguyen TT, Troendle M, Cumpston K, Rose SR, Wills BK. Lack of adverse effects from flumazenil administration: an ED observational study. *Am J Emerg Med.* 2015;33(11):1677-1679. [PubMed 26324010]

32. Patat A, Naef MM, van Gessel E, Forster A, Dubruc C, Rosenzweig P. Flumazenil antagonizes the central effects of zolpidem, an imidazopyridine hypnotic. *Clin Pharmacol Ther.* 1994;56(4):430-436. doi:10.1038/clpt.1994.157 [PubMed 7955804]
33. Penninga EI, Graudal N, Ladekarl MB, Jürgens G. Adverse events associated with flumazenil treatment for the management of suspected benzodiazepine intoxication--a systematic review with meta-analyses of randomised trials. *Basic Clin Pharmacol Toxicol.* 2016;118(1):37-44. doi:10.1111/bcpt.12434 [PubMed 26096314]
34. Proudfoot AT. Antidotes: benefits and risks. *Toxicol Lett.* 1995;82-83:779-783. [PubMed 8597142]
35. Razavizadeh AS, Zamani N, Ziaeefer P, Ebrahimi S, Hassanian-Moghaddam H. Protective effect of flumazenil infusion in severe acute benzodiazepine toxicity: a pilot randomized trial. *Eur J Clin Pharmacol.* 2021;77(4):547-554. doi:10.1007/s00228-020-03031-7 [PubMed 33125517]
36. Refer to manufacturer's labeling.
37. Reisner-Keller LA, Pham Z. Oral Flumazenil in the Treatment of Epilepsy. *Ann Pharmacother.* 1995;29(5):530-531. [PubMed 7655138]
38. Richard P, Autret E, Bardol J, et al. The Use of Flumazenil in a Neonate. *J Toxicol Clin Toxicol.* 1991;29(1):137-140. [PubMed 2005661]
39. Roald OK, Dahl V. Flunitrazepam Intoxication in a Child Successfully Treated With the Benzodiazepine Antagonist Flumazenil. *Crit Care Med.* 1989;17(12):1355-1356. [PubMed 2512053]
40. Schult RF, Omar D, Wiegand TJ, Gordetsky RM, Acquisto NM. Experience with lower dose flumazenil at an academic medical center. *Am J Emerg Med.* 2021;49:399-401. doi:10.1016/j.ajem.2021.02.002 [PubMed 33581937]
41. Sheno RP, Timm N; Committee on Drugs; Committee on Pediatric Emergency Medicine. Drugs used to treat pediatric emergencies. *Pediatrics.* 2020;145(1):e20193450. doi:10.1542/peds.2019-3450 [PubMed 31871244]
42. Spivey WH. Flumazenil and Seizures: Analysis of 43 Cases. *Clin Ther.* 1992;14(2):292-305. [PubMed 1611650]
43. Sugarman JM, Paul RI. Flumazenil: a review. *Pediatr Emerg Care.* 1994;10(1):37-43. [PubMed 8177806]
44. Trujillo MH, Guerrero J, Fragachan C, et al. Pharmacologic Antidotes in Critical Care Medicine: A Practical Guide for Drug Administration. *Crit Care Med.* 1998;26(2):377-391. [PubMed 9468179]
45. Veiraiah A, Dyas J, Cooper G, Routledge PA, Thompson JP. Flumazenil use in benzodiazepine overdose in the UK: a retrospective survey of NPIS data. *Emerg Med J.* 2012;29(7):565-569. doi:10.1136/emj.2010.095075 [PubMed 21785147]
46. Weinbroum AA, Flaishon R, Sorkine P, Szold O, Rudick V. A risk-benefit assessment of flumazenil in the management of benzodiazepine overdose. *Drug Saf.* 1997;17(3):181-196. doi:10.2165/00002018-199717030-00004 [PubMed 9306053]
47. Weinbroum A, Rudick V, Sorkine P, et al. Use of flumazenil in the treatment of drug overdose: a double-blind and open clinical study in 110 patients. *Crit Care Med.* 1996;24(2):199-206. doi:10.1097/00003246-199602000-00004 [PubMed 8605789]
48. Williams JS, Lee MA, Schauben J, et al. Flumazenil revisited. *Ann Emerg Med.* 2008;51(4):473.
49. Yang CC, Deng JF. Utility of flumazenil in zopiclone overdose. *Clin Toxicol (Phila).* 2008;46(9):920-921. doi:10.1080/15563650802043660 [PubMed 18788000]
50. Zaw W, Knoppert DC, da Silva O. Flumazenil's Reversal of Myoclonic-Like Movements Associated With Midazolam in Term Newborns. *Pharmacotherapy.* 2001;21(5):642-646. [PubMed 11349753]

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